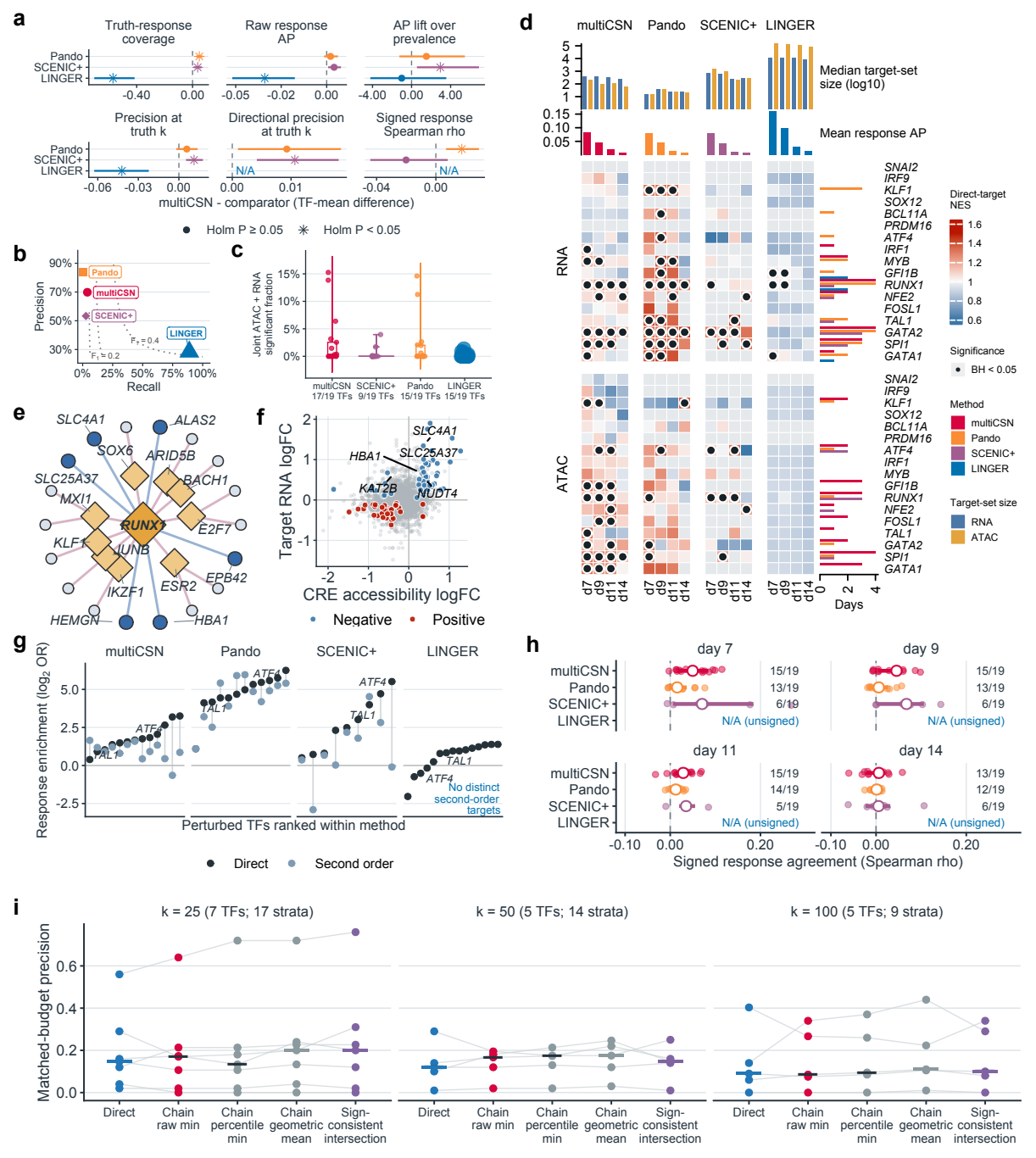

a
Truth-response coverage
Raw response AP
AP lift over prevalence
Precision at truth k
Directional precision at truth k
Signed response Spearman rho
Pando
SCENIC+
LINGER
multiCSN - comparator (TF-mean difference)
Holm P ≥ 0.05
Holm P < 0.05
b
Precision
Recall
multiCSN
c
Joint ATAC + RNA significant fraction
multiCSN 17/19 TFs
SCENIC+ 9/19 TFs
Pando 15/19 TFs
LINGER 15/19 TFs
d
multiCSN
Pando
SCENIC+
LINGER
Median target-set size (log10)
Mean response AP
RNA
ATAC
Direct-target NES
Significance
BH < 0.05
Method
Target-set size
Days
e
f
Target RNA logFC
CRE accessibility logFC
Negative
Positive
g
Response enrichment (log2 OR)
Perturbed TFs ranked within method
No distinct second-order targets
Direct
Second order
h
day 7
day 9
day 11
day 14
N/A (unsigned)
Signed response agreement (Spearman rho)
i
k = 25 (7 TFs; 17 strata)
k = 50 (5 TFs; 14 strata)
k = 100 (5 TFs; 9 strata)
Matched-budget precision
Direct
Chain raw min
Chain percentile min
Chain geometric mean
Sign-consistent intersection